

Affective Disorders

MAJOR DEPRESSIVE DISORDER
& THE ANTIDEPRESSANTS

Major Depressive Disorder (MDD) and Antidepressant (AD) Medications

Neurobiology of depression

- The neurogenic hypothesis

AD medications (e.g. TCA, SSRI, SNRI, NRI)

Role of 5-HT_{1A} autoreceptor down-regulation

Ketamine

A MAN WALKS INTO A DOCTOR'S OFFICE

- In the office of the psychiatrist, a patient is sitting in front of the doctor. This is the first time they meet and, after 30 minutes of interview, the doctor formulates a diagnosis of major depression after detecting many depressive symptoms including insomnia. At this point the doctor starts the discussion about the prescription of an antidepressant:

Serretti, A. (2018)

A MAN WALKS INTO A DOCTOR'S OFFICE

- Physician: "I understand that you are feeling a lot of suffering from your depressive condition, and I believe that an appropriate pharmacologic treatment would help you."
- Patient: "Thanks doctor, **please prescribe me the most effective antidepressant**. However, consider that I heard that many drugs may induce weight gain, and I'm very worried about it. Further, given the needs of my daily work as a driver, I would prefer drugs not causing somnolence".

Serretti, A. (2018)

A MAN WALKS INTO A DOCTOR'S OFFICE

- Bupropion (Wellbutrin) may be a good fit / high tolerability
 - Low incidence of weight gain, somnolence
 - But Px previously reported trouble with insomnia
- Citalopram (Celexa) and escitalopram (Lexapro) offer a favorable profile on weight gain, no somnolence but not increasing insomnia.

Serretti, A. (2018)

A MAN WALKS INTO A DOCTOR'S OFFICE

- Patient: "Also I forgot to tell you that 2 years ago I suffered from a myocardial infarction, and my cardiologist told me that there are still some electrocardiographic abnormalities, since then I am taking Plavix 75 mg."

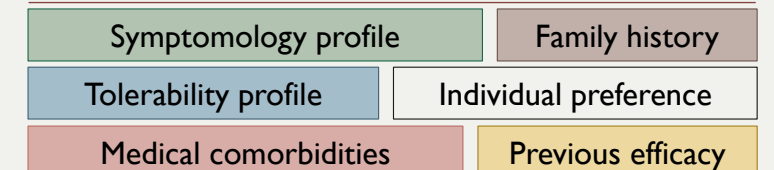
Serretti, A. (2018)

A MAN WALKS INTO A DOCTOR'S OFFICE

- Both citalopram (Celexa) and escitalopram (Lexapro) warn against use in patients with cardiac diseases.
- Both fluoxetine (Prozac) and sertraline (Zoloft) seem to have the profile needed in this specific clinical case, however
 - Drug interactions between fluoxetine and clopidogrel (Plavix) decrease the efficacy of clopidogrel.

Serretti, A. (2018)

A MAN WALKS INTO A DOCTOR'S OFFICE



- Doctor: "Taking into consideration the clinical profile of your depression, your preferences and your medical comorbidities, I think that the best option is to start sertraline".

Serretti, A. (2018)

Ch 18: MDD & the Antidepressants

Neurobiology of Major Depressive Disorder (MDD)

The monoamine hypothesis (is mostly wrong)

- based mostly on mechanisms of antidepressant (AD) meds
- too simplistic, several discrepancies

The role of stress and the stress hormone, cortisol

The neurotrophic hypothesis

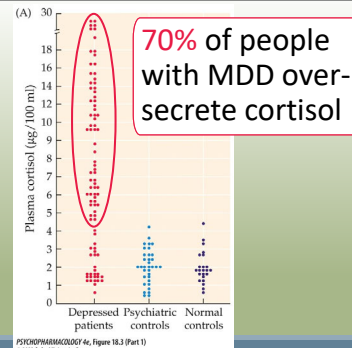
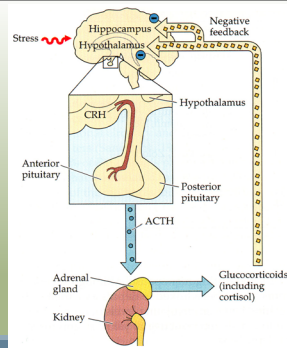
Environmental stressors activate a stress response

The stress response

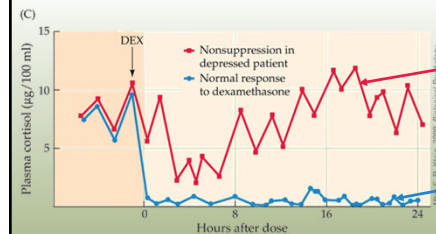
- activation of the HPA axis.
- CRH released from the hypothalamus initiates a cascade that results in cortisol release from the adrenal glands
- cortisol levels are reduced via negative feedback, involving the hippocampus
- normally follows a circadian rhythm

Environmental stressors activate a stress response

The stress response



The best-established abnormality of the HPA axis in MDD is an over-secretion of cortisol



Some people with MDD fail to suppress cortisol following dexamethasone

Blood levels of cortisol are normally reduced by dexamethasone

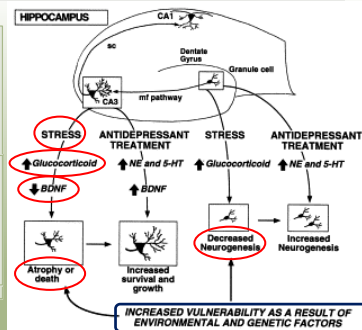
- The negative feedback system is inefficient
- The endocrine system is signaling chronic stress

Ch 18: MDD & the Antidepressants

Depression is associated with atrophy in the hippocampus.

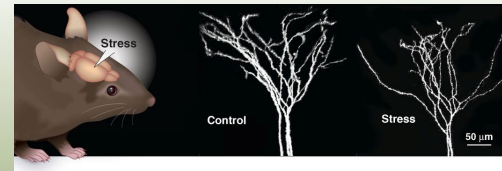
Stress increases cortisol levels, which leads to decreased BDNF

Less BDNF causes neuronal atrophy and decreased neurogenesis



Duman, R. S., Malberg, J. and Thome, J. (1999). Neural plasticity to stress and antidepressant treatment. *Biological Psychiatry*, 46, 1181 – 1191. doi:10.1016/S0006-3223(99)00177-8

Chronic stress causes atrophy of cortical neurons

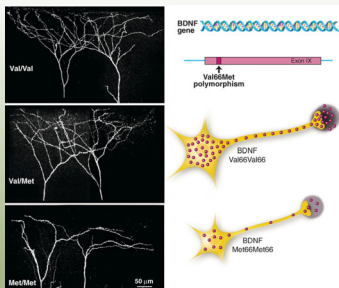


Restraint stress

reduces dendritic length and branching of mPFC neurons.

Duman, R. S. and Aghajanian, G. K. (2012). Synaptic dysfunction in depression: potential therapeutic targets. *Science*, 338, 68 – 72. doi: 0.1126/science.1222939

Chronic stress causes atrophy of cortical neurons



BDNF polymorphism, which reduces the effects of BDNF, mimics stress-induced damage.

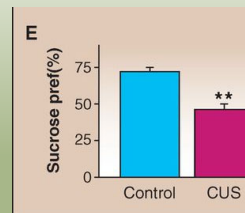
Both restraint stress and the BDNF polymorphism result in depression-like and anxiety-like behaviors in rodents

Duman, R. S. and Aghajanian, G. K. (2012). Synaptic dysfunction in depression: potential therapeutic targets. *Science*, 338, 68 – 72. doi: 0.1126/science.1222939

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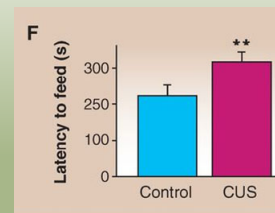
Anhedonia

Sucrose preference



Anxiety

Novelty-suppressed feeding



Duman, R. S. and Aghajanian, G. K. (2012). Synaptic dysfunction in depression: potential therapeutic targets. *Science*, 338, 68 – 72. doi: 0.1126/science.1222939

Ch 18: MDD & the Antidepressants

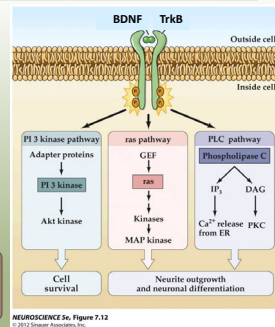
Neurotrophins are involved in depression

Chronic stress causes

elevated cortisol, which

reduces BDNF levels and activity, which leads to

neuronal atrophy & decreased neurogenesis



Remission of MDD symptoms takes time: The therapeutic lag

Antidepressants

synaptic levels of NE & 5-HT increase within hours

therapeutic effects take weeks (or months)

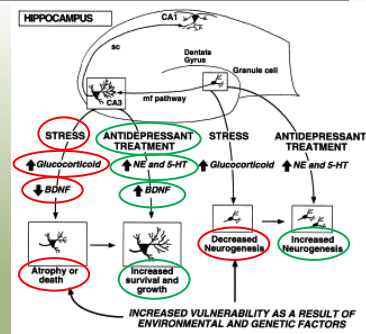
NOT the primary therapeutic mechanism.

Two alternative hypotheses

Receptor adaptations: biochemical and genetic changes that alter synaptic structure & function

AD-induced neurogenesis in the hippocampus

AD treatment promotes dendritic branching and increases neurogenesis

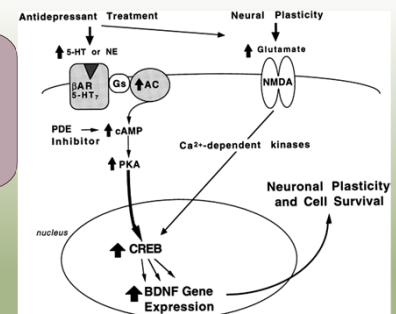


Duman, R. S., Malberg, J., and Thome, J. (1999). Neural plasticity to stress and antidepressant treatment. *Biological Psychiatry*, 46, 1181 – 1191.
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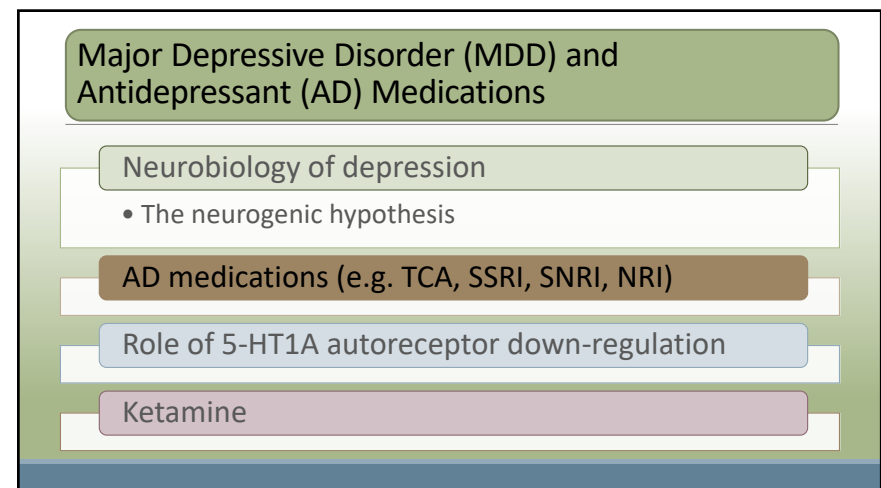
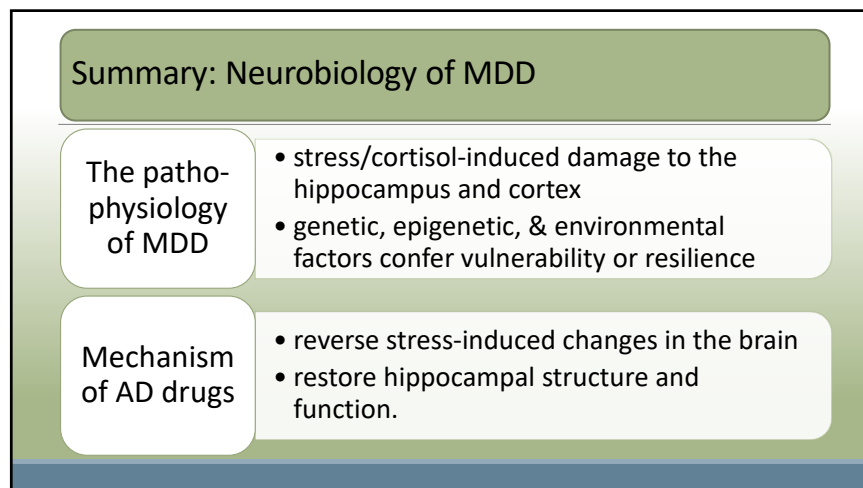
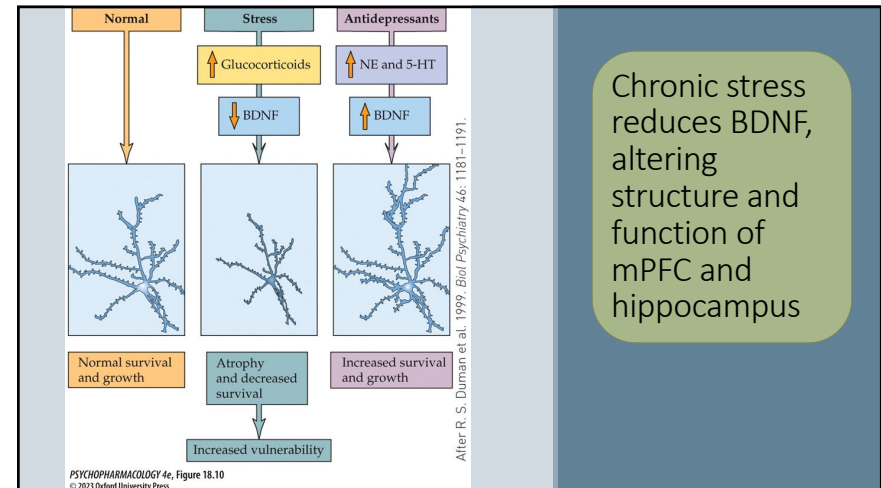
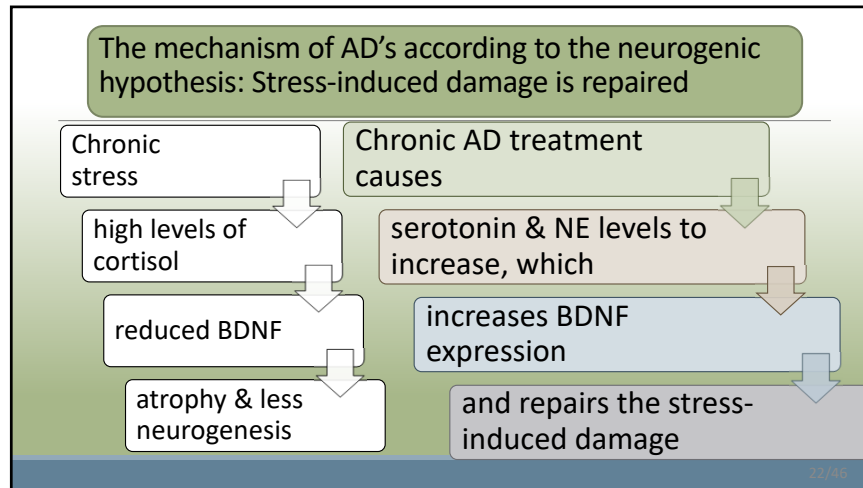
Neurogenesis in the hippocampus is proposed to be the “slow step” of therapeutic lag

- ADs increase synaptic levels of 5-HT and NE
- increased expression of BDNF



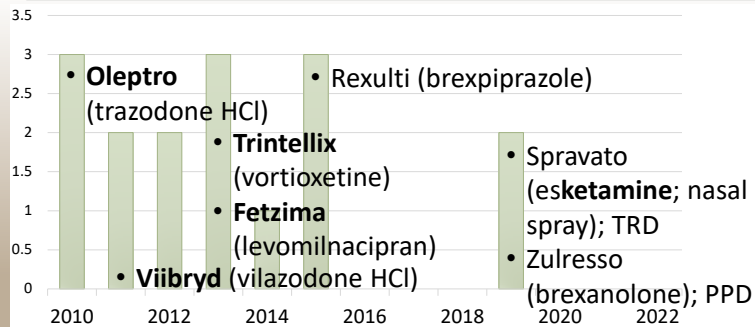
Duman, R.S., Malberg, J., Nakagawa, S., D'Sa, C. (2000). Neuronal plasticity and survival in mood disorders. *Biological Psychiatry*, 48, 732 – 739.
doi: 10.1016/S0006-3223(00)00935-5

Ch 18: MDD & the Antidepressants



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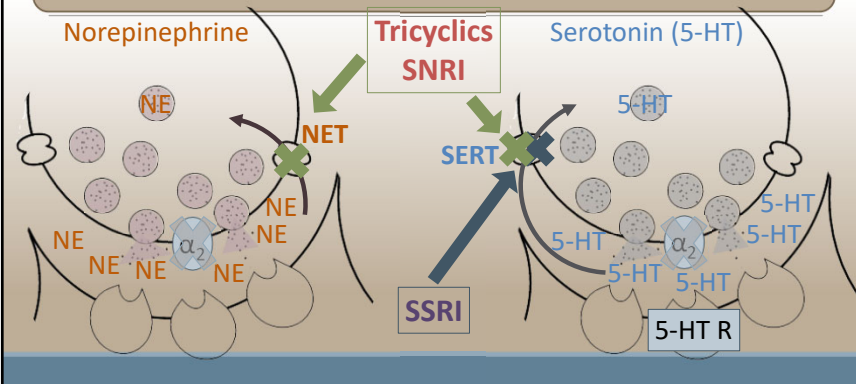
FDA-Approved Psychiatric Medications, by Year



The major classes of antidepressant (AD) drugs

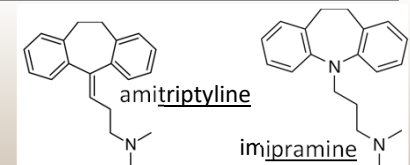
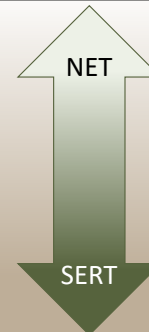
- Monoamine oxidase inhibitor (MAO-I)
- Tricyclic antidepressant (TCA)
- Selective serotonin reuptake inhibitor (SSRI)
- New generation AD's (Dual-action, Multi-action)
- Selective norepinephrine reuptake inhibitor (NRI)

Pharmacological actions of antidepressants



The specific action of tricyclic antidepressants (TCAs) is to inhibit NET and SERT

desipramine (Norpramin)
 protriptyline (Vivactil)
 amitriptyline (Elavil)
 imipramine (Tofranil)
 clomipramine (Anafranil)

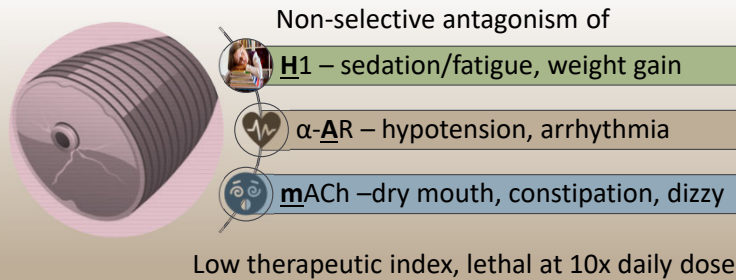


Block the reuptake of NE and 5-HT

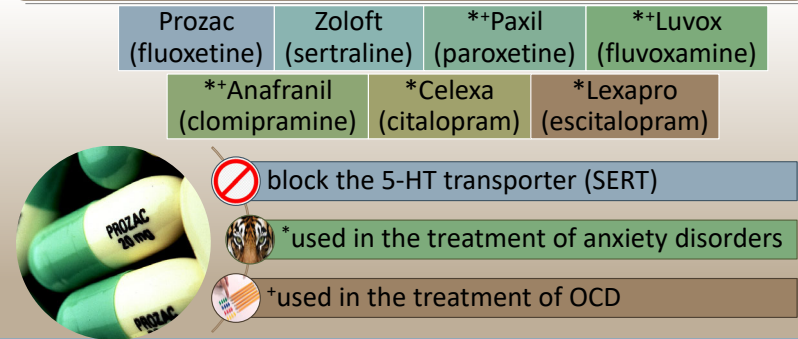
- differing affinity for SERT or NET
- unrelated to AD efficacy

Ch 18: MDD & the Antidepressants

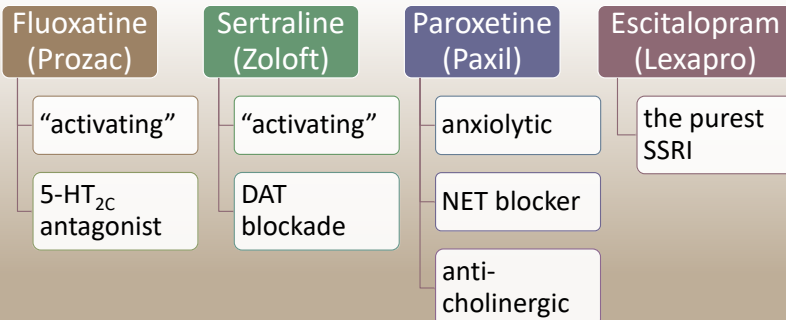
Antagonism of off-target sites causes the HAM side-effects of TCAs



The specific action of SSRI AD's is to inhibit SERT



SSRI's are distinguished by secondary actions



The side effects of SSRIs are mainly due to activation of multiple 5-HT receptor subtypes.

SSRI's are *first-line treatments* because the side-effects are generally less severe than TCAs & generally not lethal.



nausea, anxiety, insomnia, restlessness, headache



frequently (>60%) decreased libido and delayed orgasm, sometimes (~30%) anorgasmia and arousal difficulties

Ch 18: MDD & the Antidepressants

Other considerations when using SSRIs

Serotonin syndrome

- Potentially life-threatening
- agitation, ataxia, clonus, diaphoresis, tremor, disorientation, confusion
- ANS activity: fever, chills, elevated bp / hr

Discontinuation syndrome

- Dose is gradually tapered off to avoid withdrawal
- dizziness, nausea, diarrhea, fatigue, insomnia, increased anxiety, irritability

These multi-action AD's are used for their sedative properties

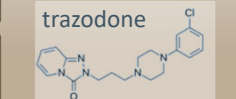
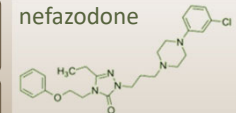
Oleptro/Desyrel (trazodone)

Serzone (nefazodone)

Serotonin Antagonist & Reuptake Inhibitor (SARI)
Blocks SERT & 5-HT_{2A} & 5-HT_{2C} receptors

Sedating (H1); effective in treatment of depression- and anxiety-related insomnia

5-HT receptor antagonism reduces SSRI-like side-effects (priapism possible)



The SNRIs are dual NE/5-HT modulators, blocking both NET and SERT

Serotonin & Norepinephrine Reuptake Inhibitors (SNRIs)

Effexor (venlafaxine)

Selectively boosts DA in PFC

side effects resemble SSRIs > TCA

Cymbalta (duloxetine)

Anti-depressant in absence of pain

Analgesic in the absence of depression

Fetzima (levomilnacipran)

FDA approved in 2013

These 2 dual-action AD's have the lowest incidence of sexual dysfunction side effects.

Wellbutrin (bupropion)

NE & DA Reuptake Inhibitor (NDRI)
cocaine-like action, but not reinforcing

Activating / stimulating, generally without sexual side effects of SSRIs

Remeron (mirtazapine)

blocks α_2 -AR to increase release of NE & 5-HT

blocks 5-HT₂ & 5-HT₃ receptors to reduce sexual side-effects

SNRI + mirtazapine = "California rocket fuel"

Ch 18: MDD & the Antidepressants

The 2 newest multi-action AD's have 5-HT_{1A} receptor *partial agonist* activity

Viibryd (vilazodone)

SPARI: 5-HT_{1A} *partial agonist* & reuptake inhibitor

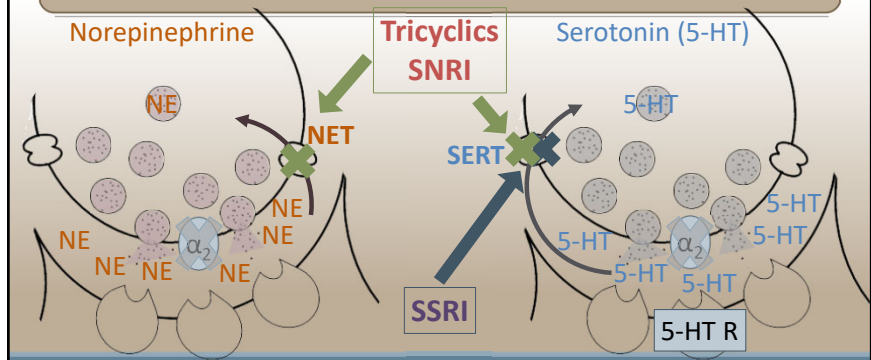
increased DA release, and lesser degree of sexual dysfunction

Trintellix (vortioxetine)

Targets include:

- SERT antagonist
- 5-HT_{1A} *partial agonist*
- 5-HT₃ antagonist
- many others (NE, DA, ACh, H)

Pharmacological actions of antidepressants



Newly approved (2019) antidepressants

Zulresso (brexanolone) for PPD

- neuroactive steroid
- positive modulator of GABA_A receptors
- continuous IV infusion over 60 hours
- headache, dizziness, somnolence

Spravato (esketamine) for TRD

- Nasal spray (used in conjunction with oral AD)
- Available only through the Spravato REMS

Major Depressive Disorder (MDD) and Antidepressant (AD) Medications

Neurobiology of depression

- The neurogenic hypothesis

AD medications (TCA, SSRI, SNRI, multi-action)

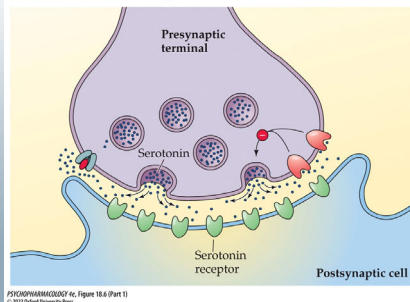
Role of 5-HT_{1A} autoreceptor down-regulation

Ketamine

Ch 18: MDD & the Antidepressants

Acute AD administration reduces release of 5-HT

(A) Acute effects of antidepressants



Acute effects of antidepressants

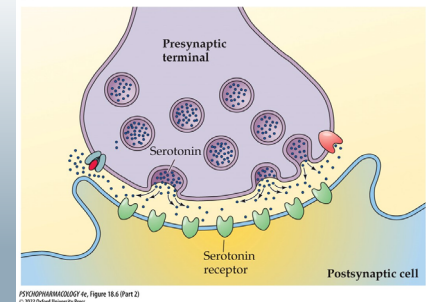
- SERT blockade elevates synaptic 5-HT
- 5-HT activates 5-HT_{1A} autoreceptors
- slows cell firing & reduces 5-HT synthesis & release

Chronic AD administration results in down-regulation of 5-HT_{1A} autoreceptors

Chronic effects of antidepressants

- 5-HT_{1A} autoreceptors down-regulate (internalize)
- 5-HT release gradually increases
- therapeutic lag?

(B) Chronic effects of antidepressants



Is 5-HT_{1A} receptor down-regulation that important?

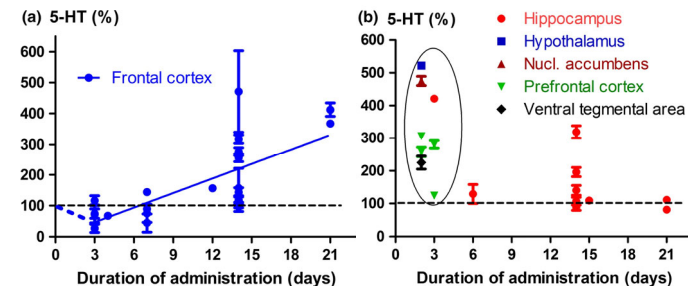
Many receptors down-regulate

- 5-HT_{1A} & α₂-AR autoreceptors
- 5-HT_{2A} & β-AR postsynaptic receptors

The time course of changing 5-HT levels depends on brain area

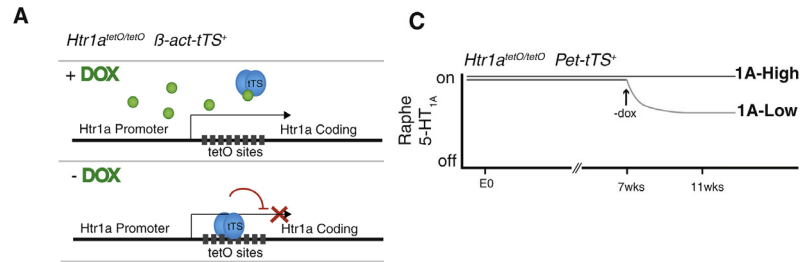
- Frontal cortex: ↓5-HT at 1 wk post-AD then ↑5-HT at 2-3 wk
- PFC, Hippo, VTA/Nacc: ↑5-HT in 3 days

The time course of changing 5-HT levels depends on brain area



Fritze, S., Spanagel, R., and Noori, H. R. (2017). Adaptive dynamics of the 5-HT systems following chronic administration of selective serotonin reuptake inhibitors: a meta-analysis. *Journal of Neurochemistry*, 142, 747-755. doi: 10.1111/jnc.14114

Is 5-HT_{1A} receptor down-regulation that important?



Richardson-Jones, J.W., Craig, C.P., Guind, B.P., Stephen, A., Metzger, K.L., Kung, H.F. ... Leonardo, E.D. (2010). 5-HT_{1A} autoreceptor levels determine vulnerability to stress and response to antidepressants. *Neuron*, 65, 40-52. doi: 10.1016/j.neuron.2009.12.003

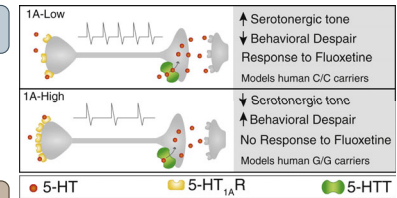
Is 5-HT_{1A} receptor down-regulation that important?

Low autoreceptor

- Faster onset (8d) of SSRI-induced changes in raphe firing rate & increased 5-HT release

High autoreceptor

- Slower onset (26d) of SSRI-induced changes in raphe firing rate and 5-HT release



Richardson-Jones, J.W., Craig, C.P., Guind, B.P., Stephen, A., Metzger, K.L., Kung, H.F. ... Leonardo, E.D. (2010). 5-HT_{1A} autoreceptor levels determine vulnerability to stress and response to antidepressants. *Neuron*, 65, 40-52. doi: 10.1016/j.neuron.2009.12.003

Is 5-HT_{1A} receptor down-regulation that important?

Resilient/responsive

- Faster onset (8d) of SSRI-induced changes in raphe firing rate & increased 5-HT release

Vulnerable/Non-responsive

- Slower onset (26d) of SSRI-induced changes in raphe firing rate and 5-HT release

"5-HT_{1A} autoreceptor desensitization alone is not sufficient for the response to fluoxetine, as 1A-High mice [show desensitization] but do not respond behaviorally to chronic fluoxetine treatment"

Richardson-Jones, J.W., Craig, C.P., Guind, B.P., Stephen, A., Metzger, K.L., Kung, H.F. ... Leonardo, E.D. (2010). 5-HT_{1A} autoreceptor levels determine vulnerability to stress and response to antidepressants. *Neuron*, 65, 40-52. doi: 10.1016/j.neuron.2009.12.003

Monoamine nuclei are highly interconnected and physiologically integrated

The "5-HT – NE hypothesis"

LC neurons alter activity of raphe

- NE increases excitability of raphe neurons
- NE increases 5-HT output; Low NE reduces 5-HT output

Raphe neurons alter activity of LC

- Overall effect of 5-HT is to inhibit (indirectly) LC output

Ch 18: MDD & the Antidepressants

Summary: The role of autoreceptors

Acute AD

↓ 5-HT synthesis & release

Worsen short-term symptoms. Suicide?

Chronic AD

down-regulation of many receptors

normal compensatory change or necessary for AD response?

explain the therapeutic lag?

Role of autoreceptor down-regulation

is complex

Multiple mechanisms involved in slow onset of AD drug effects.

Ketamine is antidepressant

- a club drug also known as “special K”
- non-competitive NMDA receptor antagonist
- effects similar to PCP
- dissociative anesthetic, primarily used as a veterinary anesthetic

At sub-anesthetic doses, IV ketamine produces a rapid but transient antidepressant effect

Ketamine is antidepressant

Ketamine clinical trials

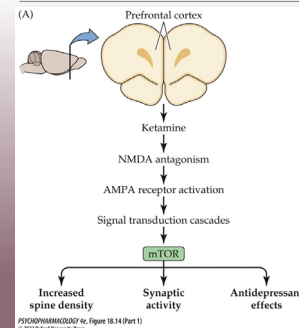
- Several small trials have shown that ketamine relieves symptoms of depression in 60 – 70% of treatment-resistant patients.

Effects are short-lived

- The antidepressant effects last only a few days, no more than 1 – 2 weeks.

The mechanism(s) of ketamine’s antidepressant effects are emerging, promising to drive new drug development

Ketamine increases spine density on dendrites in mPFC 24 hours after administration

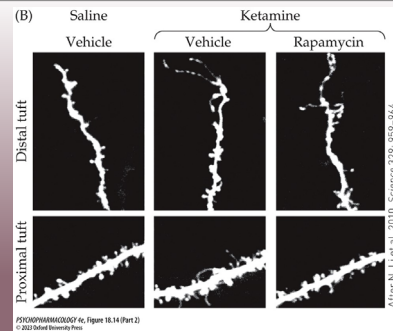


- proposed role of BDNF / TrkB
 - AD effect lost in KO mice
 - but other AD's take weeks
- rapid activation of mTOR

Ch 18: MDD & the Antidepressants

Ketamine increases spine density on dendrites in mPFC 24 hours after administration

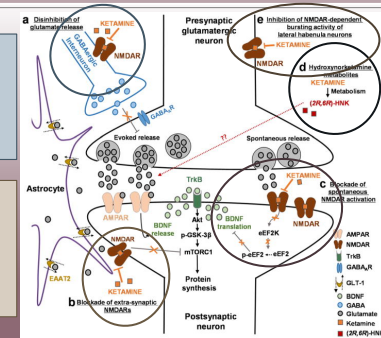
- rapid activation of mTOR
 - activation of AMPA R's
 - AD effects blocked by rapamycin
- The AD effects of ketamine may be due to a primary metabolite, hydroxynorketamine [(2R,6R)-HNK]



Proposed mechanisms of ketamine action as an antidepressant

a) selective antagonism of NMDA-R

b) antagonism of NR2B-containing NMDA-R

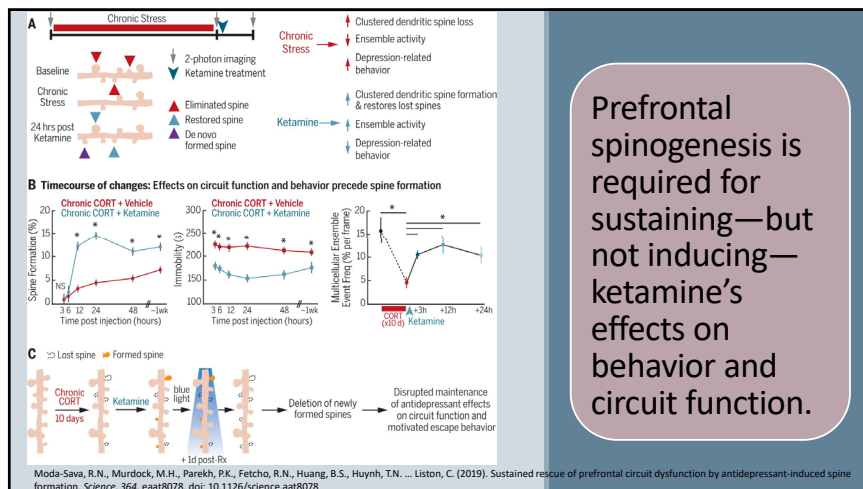


e) inhibition of habenula

d) HNK metabolites

c) NMDA-R antagonism leading to BDNF expression

Zanos, P. and Gould, T. D. (2018). Mechanisms of ketamine action as an antidepressant. *Molecular Psychiatry*, 23, 801-811. doi: 10.1038/mp.2017.255.



Prefrontal spinogenesis is required for sustaining—but not inducing—ketamine's effects on behavior and circuit function.

Synaptic remodeling in the PFC mediates the transition between depressive behavior and recovery

Stress induces, and KET rescues, clustered spine loss in mPFC and loss of ensemble activity (functional connectivity) in mPFC microcircuits.

- mPFC spinogenesis mediates transition between depressive state, remission and spontaneous recurrence of illness

Synaptic remodeling in the PFC mediates the transition between depressive behavior and recovery

Newly formed spines are not critical for behavioral recovery but are necessary for sustaining KET's AD effects over time.

- KET's AD effects occur rapidly, before new spine formation, suggesting spinogenesis is an activity-dependent adaptation to changes in circuit function.

Mechanisms likely involve BDNF/mTOR signaling in mPFC

Ketamine it is an exciting new advancement in the development of faster & more effective AD's

IV ketamine produces a rapid, but transient AD effect in animals and humans

- Mechanism is not well-defined but likely to involve restoration of stress-induced dysfunction of mPFC

Ketamine is not an ideal therapy

- clinical utility is limited by abuse potential (DEA schedule III drug) and dissociative properties